

Data Stratification, Time-Shift Augmentation, and Segmented Evaluation for ATC ASR Finetuning for Voice-Assisted Avionics

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Abstract—Voice-assisted avionics operations (VAAO) can reduce pilot workload and improve safety. However, vanilla automatic speech recognition (ASR) models cannot be used directly for air traffic control (ATC) information extraction for VAAO due to challenges such as specialized phraseology, rapid speech rates, and noise and distortions caused by band-limited radio communication. Finetuning is necessary but challenging because of variable audio quality and limited accurately transcribed data; moreover, some applications require speaker turn detection. We propose a framework with three contributions: (1) quality-based data stratification, (2) time-shift augmentation that enables turn detection and improves ASR performance on small datasets, and (3) functional transcript classification with semantic segmentation. While the first two are used for finetuning, the last is used for fine-grained segmented word error rates (WER) evaluation, which reveals that critical segments (e.g., avionics control values) achieve substantially lower WER than the conventional average WER. Results demonstrate improved finetuning and low WER on critical value segments, enabling reliable VAAO applications.

Index Terms—Voice-assisted avionics; Automatic speech recognition (ASR); ASR for Air traffic control (ATC); Quality-based data stratification; Time-shift data augmentation; Segmented word error rate (WER)

I. INTRODUCTION

A. Voice-Assisted Avionics Operations

Voice-assisted avionics operations (VAAO) refer to the application of automatic speech recognition (ASR) and natural language processing (NLP) to extract avionics operation messages and instructions from spoken communications, then prepopulate cockpit avionics systems and await pilot confirmation [1]. This capability reduces pilot workload by minimizing manual data entry and improves operational safety by allowing pilots to remain heads-up during critical flight phases.

VAAO encompasses two distinct application focuses. The first is pilot-only operations, where the pilot dictates commands directly to the avionics system (e.g., “set heading 270”). The second is ATC-based operations, where instructions from air traffic control (ATC) are automati-

cally extracted for avionics actions. This paper focuses on the latter—ATC-based VAAO—where ASR/NLP extracts action/value pairs from ATC instructions for avionics operation. The ATC-based approach can be extended to cover pilot-only cases as well; for example, after ATC instructs the pilot to “resume own navigation after waypoint OXONN,” the pilot can verbally dictate the command to the avionics when flying over OXONN, and the same underlying ASR/NLP pipeline can process that input.

The concept of VAAO is not new. A foundational patent in this area is now about a decade old [1], dating back to when Kaldi [2] was the industry-leading toolkit for ASR. Since then, numerous studies have explored assisted avionics operations [3]. However, the landscape of ASR has changed dramatically with the introduction of the Transformer [4] architecture and the emergence of end-to-end ASR models such as OpenAI’s Whisper [5] and NVIDIA NeMo # Fast Conformer [6]. ASR leaderboards [7] keeps a list of state-of-the-art models.

B. Overall Pipeline and Scope of This Paper

We are building an end-to-end ASR/NLP pipeline for ATC-based VAAO. The ultimate goal is to extract action/value pairs from ATC instructions and prepopulate avionics systems. This paper focuses exclusively on the ASR component of that pipeline. Our objective is to produce performant transcripts—accurate and speaker-turn-aware—to be consumed by the downstream NLP unit. The NLP unit, including action/value extraction and verification, is beyond the scope of this paper.

To achieve this objective, we must make a series of design choices based on our specific ATC dataset and operational constraints. This leads to the first thread of content development in this paper: a set of **research questions (Q-n)** about the best choices along our data pipeline, followed by **experimental answers (A-n)** derived from our data. While some of these questions may have been investigated by other researchers in different contexts, we ask and answer them here specifically for our data, pipeline, and target application.

Simultaneously, while answering these questions, we introduce several **new contributions (C-n)** that can be useful to other researchers working on ATC ASR or low-resource domain adaptation. These contributions form the second thread of this paper.

C. Organization of This Paper

The remainder of this paper is organized as follows. Section II discusses the ASR model selection, finetuning, and number normalization. Section III presents the ATC0 [8] dataset and introduces quality-based data stratification as a new contribution. Section IV describes time-shift augmentation for speaker turn detection and improved ASR performance as another new contribution. Section V details functional classification and semantic segmentation for fine-grained finetuning evaluation, including the rationale for segmented WER. While semantic segmentation is used extensively in NLP, it has not been used in the evaluation of ASR systems, and this is also a new contribution of this paper, as it reveals that critical segments (e.g., avionics control values) achieve substantially lower word error rates (WER) than the average WER. Section VI reports experimental results answering the research questions and showing the effectiveness of our framework. Section VII concludes with discussion and future directions. Two appendixes provide error case illustrations for action and value segments.

II. ASR MODELS, FINETUNING, AND NUMBER NORMALIZATION

A. OpenAI Whisper Models

We select OpenAI’s Whisper Large V3-Turbo [5] as the vanilla ASR model for finetuning in this work. Several factors motivate this choice:

- Strong baseline performance on general-domain speech recognition.
- Fast transcription speed suitable for real-time cockpit applications.
- Compatibility with real-time streaming frameworks such as WhisperLiveKit [9].
- Robust performance even under very low signal-to-noise ratio (SNR) conditions.
- Hallucination can be effectively suppressed when combined with voice activity detection (VAD) [10].
- Easy finetuning via Low-Rank Adaptation (LoRA) [11], which is critical given our limited ATC dataset.
- Natural output of transcripts with casing and punctuation, which aids readability and downstream NLP processing.

We acknowledge that the ASR Leaderboard lists many models with superior evaluation metrics on standard benchmarks compared to Whisper. However, after finetuning on ATC0 data, we observe that Whisper models outperform many alternatives. While we do not have a definitive explanation, we hypothesize that Whisper’s training on extremely large and diverse corpora—spanning multiple

languages and acoustic conditions—enables better generalization to the specialized ATC domain after finetuning, even if its zero-shot performance on certain test sets is not outstanding.

B. Parameter-Efficient Model Finetuning

Deploying ASR for VAAO in real cockpit environments faces substantial challenges. Standard “vanilla” ASR models perform poorly on ATC communications due to specialized phraseology, rapid speech rates, and noise and distortion from band-limited radio transmissions. Finetuning is therefore necessary.

Among finetuning approaches, LoRA [11] is one of the most popular standard approaches for parameter-efficient training. LoRA injects trainable low-rank matrices into the transformer layers while freezing the base model weights, significantly reducing memory and compute requirements—a critical advantage for resource-constrained avionics platforms.

Recent work, such as [12], have confirmed our gains due to LoRA-based finetuning to ATC ASR tasks.

Our finetuned model targets offline processing, where entire ATC transmissions are transcribed after reception. Nevertheless, the same model supports streaming applications when combined with WhisperLiveKit [9].

C. Number Normalization

Whisper models include a built-in number normalizer that converts spoken (word form) numbers into digit strings (e.g., “two four zero” -> “240”). For general ASR, this is beneficial. However, for ATC-based VAAO, we explicitly disable this normalization and retain word-based number representations. Three reasons motivate this decision.

First, we wish to preserve the exact pronunciation of numbers as spoken by controllers and pilots. While not directly required for avionics operations, this preserves the ability to evaluate pilot adherence to ATC phraseology standards—a potential training and safety monitoring tool.

Second, word-level output allows us to retain timestamps for each token. These timestamps enable downstream fluency analysis and alignment with other cockpit data streams.

Third, number normalization can produce incorrect concatenations that destroy readability. Consider the following example:

- **ATC instruction:** “Delta one twenty three, descend and maintain flight level **two four zero**, reduce speed to **two one zero**.”
- **Pilot readback:** “**Two four zero, two one zero**, Delta one twenty three.”

A conventional number normalizer would produce: “240210, Delta 123” for the pilot readback when punctuation is not transcribed or removed. This output dramatically increases the difficulty of downstream NLP tasks such as action/value extraction. By contrast, our word-based normalization preserves “two four zero” and

"two one zero" as separate token sequences, deferring numerical interpretation to the NLP unit. This approach has proven effective in our pipeline.

Note that all our WER evaluations are based on word-format number normalizer.

D. Research Questions 1 and 2

We need to ask and answer the following are research questions before proceeding:

Q-1: How will the vanilla Whisper Large V3-Turbo model perform on the ATC0 dataset without finetuning?

A-1: Poorly. The raw word error rate (WER) exceeds 60%, rendering the model unusable for VAAO applications. This establishes the clear necessity for domain-specific finetuning.

Q-2: How does LoRA adapter size affect finetuning results on our ATC0 dataset? This question will be answered later in Section VI as A-2.

III. DATASET AND QUALITY-BASED DATA STRATIFICATION

A. The ATC0 Dataset

Publicly available datasets for ATC ASR remain relatively scarce compared to general-domain speech recognition. Several corpora have been developed over the years, including ATC0 (the LDC ATC dataset) [8] and ATC02 [13]. Each presents trade-offs between size, annotation quality, speaker variability, and acoustic conditions. A comprehensive survey of ATC ASR datasets can be found in [13].

For our work, we have been using the ATC0 dataset, which consists of recorded ATC communications from three US airports, including DCA (Ronald Reagan Washington National Airport), encompassing a range of Approach, Departure, Local, and Ground control frequencies. The dataset includes both controller and pilot transmissions with corresponding transcriptions. Audio quality varies significantly across transmission due to differences in radio equipment and the relative location between transmitters and the recording receiver. For convenience of presentation, we define two terms related to a transmission:

- **Cue** if we focus on the transcript of the transmission, a term borrowed from subtitles. A cue should have its own timestamps, targeting a single listener. When ATC keys the mic to talk to multiple pilots in one transmission, the transmission will be counted as multiple cues.
- **Clip** if we focus on the audio of the transmission corresponding to a cue.

A critical limitation in ATC ASR research is the absence of a standardized, publicly accepted benchmark dataset for cross-system comparison. Consequently, we cannot directly compare our absolute WER numbers to those reported in other papers. Instead, we focus on **vertical comparisons** within our own pipeline. As a reference point, we were able

to achieve about **10%** WER by finetuning Whisper V3 Turbo on the ATC0 dataset.

B. Ground Truth Issues and Dataset Revision

During error analysis of initial finetuning experiments, we discovered that a non-trivial portion of ASR errors stemmed not from model limitations but from **ground truth transcription errors** in the original ATC0 dataset. These included errors of time stamps and mis-transcriptions.

We have since revised a small portion of the ATC0 data, covering the Approach and Departure frequencies from DCA. After retraining and evaluating on the corrected data, finetuning results improved notably. Note that the test dataset received more careful annotation review than the training dataset. We also added consistent casing and punctuation to all transcripts.

A full description of the dataset revision process, including customized tools, will be presented in a future paper once more data has been processed and reviewed. Therefore, we do not claim improved ground truth labeling as a contribution of this paper although we use the revised portion of the dataset.

C. Quality-Based Data Stratification

Despite ground truth corrections, we observed that some audio clips remain extremely difficult to transcribe accurately, even for human listeners. This is due to the inherent challenges of ATC communications: fast speaking rates as well as noise and distortion caused by band-limited radio transmissions.

Rather than discarding low-quality clips or treating all data uniformly, we decided to implement **quality-based data stratification**—a systematic labeling of audio transmissions into different audio quality levels, **intending to see if the inclusion of lower quality audio will negatively affect the transcription performance**.

We define four audio quality levels based on complete transmission (from one speaker to one listener), either from ATC or pilot.

Level 1. The audio is **somewhat clear**. The evaluator has no issues deciphering the audio to transcripts themselves if needed. The transcript matches the audio well. If mis-transcriptions are identified, they are corrected.

Level 2. The audio is **mostly intelligible**. The evaluator may have some difficulty transcribing the audio themselves, but when they see the provided transcript, they should think that **the transcript accurately reflects the audio**.

Level 3. The audio is **difficult to make out**. **Part of the provided transcript does not make much sense even after comparing the audio and transcripts a few times**. (Note, this is not the same as there is mis-transcription in the provided transcript.) There should be no unintelligible parts in the provided transcripts, and the

evaluator *chooses to trust the provided transcript*. There can be a few reasons leading to this situation:

- The volume of the audio is too low, and an amplified version is hence too noisy.
- The audio is affected by an aircraft warning audio or noise.
- The ATC or the pilot was talking extremely fast with murmured or skipped sounds.

Level 4. The audio is unintelligible and there are unintelligible parts in the provided transcripts. There is no way for the evaluator to trust the provided transcript, and there will be no value for ASR training. There are two ways to handle such a case: Label it as Level 4 or just remove it. The reason we want to keep it as Level 4 is that it could be useful for NLP, a downstream processing. The reason leading to this situation is mainly due to radio transmission blockage.

These audio quality levels are evaluated and assigned by the authors—engineering researchers with limited domain expertise in ATC communications. While we do have ATC experts on our team, the labeling in this paper was performed by engineering researchers to demonstrate that **this stratification approach can be replicated by other engineering researchers** working in ATC ASR or other low-resource domains, without requiring deep domain expertise. This replicability is an intentional design choice.

When using these level-labeled data for training and evaluation, we often put them together in a progressive way:

- **L1**: Only Level 1 (high-quality) transmissions
- **L12**: Levels 1 and 2 (high + medium)
- **L123**: All levels (high + medium + low)

D. Available Data for Subsequent Experiments

Transcript revision and quality-based data stratification require a lot of efforts, and we only have a small portion of ATC0 data processed. By design, the processed data files cover the Approach and Departure segments from the DCA facility. With the data for Approach and Departure, finetuned VAAO can cover the different ATC positions/frequencies including Departure, En Route, and Approach.

To experiment with the augmentation approaches, we separate the DCA data into training and test data.

1) *The DCA Training Data*: The DCA training data is divided into three equal sized batches, each having 120 minutes for Departure and 100 minutes for Approach. This is a total of 3.67 hours of training data.

Batch 1 files:

- dca_d1_3 , 120 min. (d1 stands for Departure frequency 1, and 3 recording file 3.)
- dca_f1_2 , 120 min (the first 100 minutes). (f1 stands for final Approach frequency 1, and 2 recording file 2.)
- End at time (seconds): 5952.530.

Batch 2 files:

- dca_d2_1 , 60 min.
- dca_d2_2 , 61 min.
- dca_f2_3 , 120 min (the first 100 minutes)
- End at time: 5987.690

Batch 3 files:

- dca_d1_1 , 90 min (the first 50 minutes)
- End at time: 2968.660
- dca_d1_2 , 70 min.
- dca_f2_1 , 60 min.
- dca_f1_2 , 120 min (the last 20 minutes).
- Start from time: 6081.480
- dca_f2_3 , 120 min (the last 20 minutes)
- Start from time: 6062.610

So, we have three sizes of training datasets, which are all relatively small.

- **B1**: Only Batch 1 (3.67 hours)
- **B12**: Batches 1 and 2 (7.33 hours)
- **B123**: Batches 1, 2, and 3 (11 hours)

2) *The DCA Test Data*: We use a relatively large test data to create a better statistics. The test data has about 270 minutes (4.5 hours):

- dca_d1_1 , 90 min (the last 40 minutes)
- Start from time: 3011.130
- dca_d1_4 , 120 min
- dca_f1_1 , 56 min.
- dca_f2_2 , 60 min.

E. Research Question 3 and Contribution 1

Q-3: Does using quality-based data stratification with progressive inclusion of different data tiers (L1, L12, L123) affect the performance of finetuned ASR models on ATC0?

A-3 (preview): Yes. The answer, along with quantitative results, will be presented in Section VI. We find that selective training on L12 data generally improves performance on all levels of transmission.

C-1: Quality-based data stratification for ATC ASR finetuning and evaluation. To the best of our knowledge, this is the first work to systematically label ATC audio transmissions by intrinsic audio quality levels and use tiered inclusion strategies for both training and evaluation. Our explicit human-annotated quality stratification and the level/tier distinction provides a replicable framework for other researchers. This contribution can be extended to other ATC datasets and potentially to other low-resource domain ASR tasks where audio quality varies widely.

Note that quality-based data stratification can be applied to both finetuning (selecting which clips to train on) and evaluation (reporting performance separately for each quality tier). We adopt both uses in this paper.

IV. TIME-SHIFT AUGMENTATION FOR FINETUNING

A. Motivation: Two Requirements for ATC ASR in VAAO

Our ultimate goal is to finetune a single ASR model that satisfies two critical requirements for ATC-based

VAAO: speaker turn detection and transcription. This unified model approach is essential because computational resources on aircraft are limited in terms of both memory and power; running separate models for speaker turn detection and transcription is often impractical.

Speaker turn detection can usually be performed by speaker diarization. Yet, we have been experiencing performance issues with the current solutions. So, we decide to do turn detection using ASR models.

Requirement 1: Speaker Turn Detection. In ATC communications, multiple parties (controllers and multiple pilots) share the same radio frequency. The ability to detect who is speaking—and when speaker turns occur—enables the VAAO system to maintain situational awareness of “party-line” information. When ATC corrects a pilot or another pilot points out a potential miscommunication, the system can capture these exchanges and alert the flight crew. The ATC0 dataset contains several real-world examples where miscommunications were caught by other parties on the frequency. Furthermore, speaker turn detection helps segment continuous audio streams into discrete transmissions, facilitating downstream NLP processing.

When comparing two ASR models—one with speaker turn detection capability and one without—we will prefer the model with turn detection as long as its transcription performance does not degrade excessively. This preference motivates our use of time-shift augmentation to bake speaker turn detection directly into the finetuned model.

Requirement 2: Improved ASR Performance. This is the primary goal of most ASR finetuning research. Lower WER directly translates to fewer incorrect action/value extractions and higher pilot trust in the VAAO system.

B. Conditions for Supporting Speaker Turn Detection

There are two conditions to support speaker turn detection by finetuning with authentic training data. The first is that the audio files of the dataset should be long form files, meaning each contains multiple ATC transmissions. The second is that there should be enough transmissions that are adjacent to each other in the audio files so these transmissions can be combined to carry speaker turn information. Fortunately, both are satisfied in ATC0.

C. Time-Shift Augmentation

Time-shift augmentation is a shortened name for “time-domain shift-based cue-stitching augmentation”. The principle is described below:

- **Time-domain:** The augmentation is performed exclusively in the time domain. (Other augmentation approaches can be applied on top of this.)
- **Cue-stitching:** In un-augmented training, we use Segment Time Marking (STM) files to label the timestamps of each transmission. Using the same approach, we can stitch two adjacent cues together

to form a *stitched cue*, enabling straightforward time-domain data augmentation.

- We insert a turn change symbol, `//`, between the two cues so that the ASR model learns to detect the speaker turn automatically. (We experimented with `||`, but that led to hallucinations.)
- The resulting concatenated sentence must remain under 30 seconds, the input length limit of Whisper models.

- **Time shift:** When a single cue is used multiple times, we apply a time shift so that the augmented data does not always start at the exact same point. This follows conventional augmentation practice.
- **Equal chance:** For transmissions occurring at the two ends of a sequence of adjacent cues, they are repeated to ensure each transmission has a fair chance of being included.

We consider two approaches for augmentation, as detailed below:

1) Augmentation Approach 1—Double Augmentation:

This approach focuses on using each cue **twice**. The idea is illustrated below:

For a block of single cue:

```
Orig: {-cue0-}
Aug1: {-cue0-}  0 ms
Aug1: {-cue0-}  5 ms
```

For a block of two adjacent cues:

```
Orig: {-cue0-} {-cue1-}
Aug1: {-cue0-}  0 ms
Aug1: {-cue0- // -cue1-}  5 ms
Aug1:                {-cue1-}  0 ms
```

For a block of three adjacent cues:

```
Orig: {-cue0-} {-cue1-} {-cue2-}
Aug1: {-cue0-}  0 ms
Aug1: {-cue0- // -cue1-}  5 ms
Aug1:                {-cue1- // -cue2-}  0 ms
Aug1:                {-cue2-}  5 ms
```

Note that :

- `Orig` is the original cue(s), and `aug1` are the new cues of the Augmentation 1, which can be hosted in a new STM file.
- 0 ms or 5 ms denotes the sampling shifts in terms of millisecond as compared to the begin time of time stamps. This way, two samples of the same cue will have 5 ms offset, which works well in the case of 10 ms sliding frames used in most ASR system. Note that we can use 15 ms instead of 5 ms, but a 10 ms will not make it.

2) Augmentation Approach 2—Triple Augmentation:

The key for Approach 2 is to use each cue **three times**:

- Non-augmented case. This for the independent training case.
- Advance 13 ms for augmenting, used for stitched cues if applicable.
- Advance 26 ms for augmenting, used for stitched cues if applicable.

For a block of single transmission:

```
Orig: {-cue0-}
Aug2: {-cue0-} 0 ms
Aug2: {-cue0-} 13 ms
Aug2: {-cue0-} 26 ms
```

For a block of two transmissions:

```
Orig: {-cue0-} {-cue1-}
Aug2: {-cue0-} 0 ms
Aug2: {-cue0-} 13 ms
Aug2: {-cue0- // -cue1-} 26 ms
Aug2: {-cue1-} 0 ms
Aug2: {-cue1-} 13 ms
```

For a block of three transmissions:

```
Orig: {-cue0-} {-cue1-} {-cue2-}
Aug2: {-cue0-} 0 ms
Aug2: {-cue0-} 13 ms
Aug2: {-cue0- // -cue1-} 26 ms
Aug2: {-cue1-} 0 ms
Aug2: {-cue1- // -cue2-} 13 ms
Aug2: {-cue2-} 26 ms
Aug2: {-cue2-} 0 ms
```

We can see that the major difference between Approaches 1 and 2 is that the latter use more single transmissions.

D. Research Questions 4 and 5 and Contribution 2

Q-4: Will speaker turn detection work effectively with models finetuned using turn-aware concatenation augmentation?

A-4 (preview): Yes. As will be shown in Section VI, the finetuned model reliably outputs // tokens at speaker boundaries, achieving high precision and recall on test data containing multiple turns.

Q-5: How does time-shift augmentation affect WER performance, particularly when training data size is limited?

A-5 (preview): Augmentation improves WER when the effective training data size is small. For our full dataset, the improvement is not noticeable.

C-2: Time-shift augmentation for joint ASR and speaker turn detection in ATC applications. To the best of our knowledge, this is the first work to use concatenation-based augmentation with separator tokens to bake speaker turn detection into an end-to-end ASR model finetuned for ATC. While prior work has focused on speaker diarization separately or used transcript-based processing for turn detection [14], our approach trains a single model to output both transcription and turn boundaries simultaneously. This

is particularly valuable for resource-constrained cockpit environments where running separate models is prohibitive. Furthermore, our augmentation strategy improves WER in low-data regimes, addressing a common challenge in ATC ASR.

V. FUNCTIONAL CLASSIFICATION AND SEMANTIC SEGMENTATION FOR EVALUATION

A. Motivation: Non-Uniform WER Requirements for VAAO

A VAAO system does not require uniform ASR performance across all words of an ATC transmission. Consider an instruction such as: “Delta one twenty three, reduce speed to two one zero. Good day.” For downstream avionics operations, misrecognizing “Good day” will not affect the VAAO. Therefore, evaluating ASR performance solely via overall WER can be misleading. A model that achieves, for example, 5% overall WER might have 1% WER on critical value segments (acceptable) or 15% WER on critical value segments (unacceptable). The average WER number obscures this distinction.

This observation motivates our development of **segmented WER evaluation** based on **functional classification** and **semantic segmentation**.

B. Related Work on Segmented and Intent-Based ASR Evaluation

To the best of our knowledge, [15] is the work that is most closely related to our segmented WER evaluation. Yet, it is on the word level, such as “keyboard” vs “key board”, and we work on the semantic level, as shown below.

Our work for functional classification and semantic segmentation is influenced by [16] and [17], and we propose a much more simpler, more accessible method: functional classification and semantic segmentation.

C. Classifying ATC Transmissions into Functional Classes

We classify ATC transmissions into the following **four functional classes** according to their nature.

- (1) **ATC-IA.** ATC transmissions containing instructions for immediate avionics operations (e.g., heading or altitude changes).
- (2) **PLT-RA.** Pilot readbacks that correspond to ATC-IA messages.
- (3) **ATC-OT.** Other ATC transmissions, including future-phase instructions (e.g., speed constraints to a certain waypoint), acknowledgments, and routine exchanges. There is no pressure to extract future-phase instructions as pilots will have enough time for these actions.
- (4) **PLT-OT.** Other pilot transmissions, including readbacks for non-avionics instructions, requests, and routine exchanges.

D. Semantic Segmentation of Each ATC Transmission

We consider the following semantic segments, which have different levels of correlation to action/value pair extraction for assisted avionics operation.

A. **Callsign.**

B. **Action.** A verb phrase indicating the instructed/intended avionics operation (e.g., “climb,” “descend,” “turn”) in ATC-IA or PLT-RA.

C. **Value.** Numeric parameters of an *action*, which includes units in ATC-IA or PLT-RA. For action and value, we can further divide them into types: 1) altimeter, 2) altitude, 3) frequency, 4) heading, 5) speed, 6) squawk.

D. **Acknowledge.** Short confirmations like “roger,” “wilco,” “affirm,” “will do”, and “radar contact” in all above classes.

E. **Info.** Expressions used for providing info only, such as “going from 4 to 5” if not in the required readback. If it is in readback, it will be action and value.

F. **Condition.** Condition phrases indicating when an action should occur (e.g., “after passing,” “when able”) in ATC-IA or PLT-RA.

G. **Facility.** Identifiers for various facilities that do not have their unique name (except city names). These include such ATC position, runway, ILS, and DME.

H. **Navaid.** Named geographic points (e.g., “JFK VOR,” “ALBANY”) used in all above classes.

I. **Greeting and farewell.** Salutations such as “good morning”, “good day”, “hello”, and “good bye” in all above classes.

J. **Other.** Everything that cannot be classified into the above segments.

Below are some examples:

```
-----
stm: 1>D1-2>UAL619
tim: 3327350
txt: United Six Nineteen, contact Dulles
      Departure one three four point two.
LLM: (1) [(A) United Six Nineteen] [(B3)
      contact] [(G) Dulles Departure]
      [(C3) one three four point two]

stm: 2>UAL619>D1-2
tim: 3331230
txt: Thirty four two, good day, United
      Six Nineteen.
LLM: (2) [(C3) Thirty four two] [(I) good
      day] [(A) United Six Nineteen]
-----
```

Note that:

- The `stm` (STM) layer records the audio quality level, the speaker, and the receiver.
- The `tim` (time) layer records the begin time of the transmission in the units of milliseconds.
- The `txt` (text) layer provides the verbatim transcript.

- The LLM layer is the LLM-generated categorized segmentation which is revised by human reviewers and checked by customized tool.
- Categorized segments can be nested, like [(E) turn right direct to [(H) LINDEN]] to reduce the labeling effort and increase the flexibility of processing. For example, we can choose to separate the (E) and (H) categories or can merge (H) into (E).

E. Segmented WER Evaluation

A literature search revealed no existing tools for segmented WER evaluation, likely due to the limited use of semantic segmentation in ASR. We therefore developed our own tools, which will be fully described in a future paper. The tooling itself is not claimed as a contribution of this work.

Note that we have the following modifications to the transcripts when doing segmented WER evaluation:

- We have removed all filler pauses, which was normalized to “er” during the finetuning. They are not used in our NLP pipeline anyway.
- We have removed all punctuations and casing for the ASR transcripts and the ground truths for a easy comparison. They are not needed in our NLP pipeline either.

F. Creation of Semantic Segments for the Ground Truth Data

Our segmented WER evaluation tooling requires only the labeling of functional classification and semantic segmentation on the ground truth data. We used a three-step process to create this labeling:

- **LLM-based initial labeling:** Process each transcription, along with its history context, using a large language model (LLM) to produce an initial categorization and segmentation.
- **Manual review and correction:** Manually review and correct each result to ensure accuracy.
- **Rule-based validation:** Run a rule-based tool to verify that the segments correctly match the transcripts. This step is primarily used to detect LLM hallucinations and manual editing errors. While some errors may be tolerable for ASR finetuning, they must be completely eliminated here because this part is crucial—errors in segmentation directly determine the accuracy of segmented WER calculation.

G. Contribution 3

C-3: Functional classification and semantic segmentation for fine-grained ATC ASR evaluation. While prior work has explored intent-based evaluation with complex ontologies, our approach is deliberately lightweight and accessible to engineering researchers without deep ATC expertise. As we will see later in performance evaluation, we can see Value segments achieve lower WER than average—directly informs the design of VAAO systems.

Table I: WER of the vanilla Whisper V3 Turbo model on the test dataset.

L1	L12	L123
66.22%	66.13%	66.27%

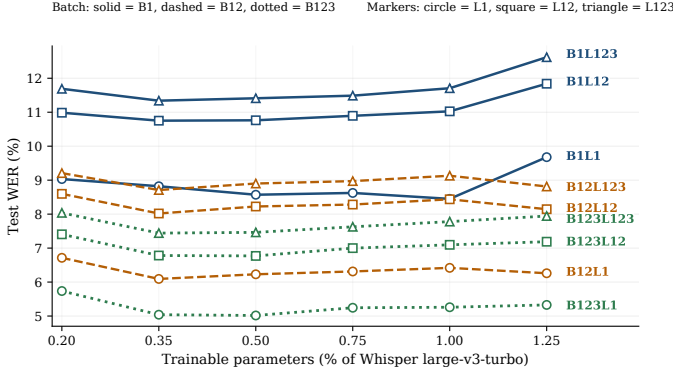


Figure 1: WER of different cases as a function of model size for normal finetuning.

VI. PERFORMANCE EVALUATION

A. Baseline WER with Vanilla V3-Turbo

Before we preset the experiments using finetuning, let’s look at performance of the vanilla Whisper Large V3 Turbo model, as shown in Table I.

B. WER as a Function of LoRA Model Size Without Augmentation

Now, we examine finetuning performance as a function of LoRA adapter size to identify appropriate hyperparameters, prior to applying any augmentation. The following parameters and data configurations were used:

- **LoRA sizes:** 0.2%, 0.35%, 0.5%, 0.75%, 1.0%, and 1.25%.
- **Data levels for finetuning:** L12.
- **Training data batches:** B1, B12, and B123.
- **Evaluation data:** L1, L12, and L123.

Figure-1 presents the WER results across these configurations. Based on these results, a LoRA size 0.5% is a good choice for the amount of training data used in this study. For substantially larger datasets, we predict 1% or 1.25% may become more appropriate.

1) *Answer to Research Question 2: Q-2:* How will LoRA size affect the finetuning results on **our data**? Will be answered later as A-2.

A-2: The optimal LoRA adapter size depends slightly on the training dataset size. For the amount of data of this study (approximately 11 hours), a LoRA size of 0.5% of the base model is an appropriate choice.

C. WER as a Function of Audio Quality Without Augmentation

With the LoRA size fixed as 0.5%, we next finetune models with different levels of training data: L1, L12, and L123. Again, no augmentation is applied here.

We also evaluate the WER on different audio quality levels, as shown in Table II.

1) *Answer to Research Question 3: Q-3:* Does using quality-based data stratification with progressive inclusion of different data tiers (L1, L12, L123) affect the performance of finetuned ASR models on ATC0?

A-3: We find that selective training on L12 data generally improves performance on all levels of transmission.

The likely explanation is twofold. First, training on L1 alone provides insufficient data quantity, limiting generalization. Second, training on L123 introduces incorrectly labeled or highly degraded samples from Level 3, which can negatively affect model performance. Training on L12 strikes an effective balance: sufficient data volume without the harmful influence of the lowest-quality samples.

D. WER as a Function of Different Augmentations

With LoRA size fixed at 0.5% and training data fixed at L12, we now examine the impact of different augmentation approaches on finetuning performance. Evaluation is again performed separately on each audio quality level (L1, L12, L123).

We compare three augmentation configurations: (1) No augmentation (baseline), (2) Augmentation approach 1, and (3) Augmentation approach 2.

Table-III presents the WER results across these configurations and evaluation tiers.

1) *Answer to Research Question 5: Q-5:* How does time-shift augmentation affect WER performance, particularly when training data size is limited?

A-5: Augmentation improves WER when the effective training data size is small. For our full dataset, the improvement is not noticeable. At least there is no WER performance degradation with the turn-aware models.

E. Segmented WER for VAAO

Segmented WER is particularly well-suited for VAAO applications because errors in certain categories do not affect the extraction of action/value pairs. For the experiments in this section, we use the following configuration:

- **Training data:** L12.
- **Finetuned models:** Two variants—one trained on B1 and the other trained on B123.
- **Evaluation data:** L12 only.

1) *WER on Semantic Segments Involved in Action/Value Pair Extraction:* When ATC provides instructions, Class 1, we are primarily interested in three semantic segments: Seg A (callsign), Seg B (action), and Seg C (value).

Table IV presents segmented WER results for these segments. For each segment category, the values in parenthesis following the category name are the total number of words

Table II: Raw WER comparison using different quality data for finetuning.

	B1:L1	B1:L12	B1:L123	B12:L1	B12:L12	B12:L123	B123:L1	B123:L12	B123:L123
L1	9.33%	11.92%	12.52%	6.28%	8.19%	9.51%	5.35%	7.49%	8.21%
L12	8.57%	10.76%	11.41%	6.22%	8.22%	8.90%	5.01%	6.77%	7.46%
L123	8.71%	11.49%	12.05%	7.04%	8.83%	9.43%	4.78%	6.55%	7.20%

Table III: Raw WER comparison using different augmentation approaches for finetuning.

	B1:L1	B1:L12	B1:L123	B12:L1	B12:L12	B12:L123	B123:L1	B123:L12	B123:L123
No aug	8.57%	10.76%	11.41%	6.22%	8.22%	8.90%	5.01%	6.77%	7.46%
Aug 1	8.44%	10.70%	11.36%	6.03%	7.91%	8.57%	4.94%	6.83%	7.57%
Aug 2	7.99%	10.40%	11.05%	5.70%	7.74%	8.50%	5.27%	7.17%	7.83%

Table IV: WER on segments used for action/value pair extraction.

	Seg A (2514)	Seg B (1508)	Seg C (2220)
B 1	10.700%, 269	4.708%, 71	2.928%, 65
B 123	5.648%, 142	3.647%, 55	0.541%, 12

Table V: WER on segments used for action/value pair matching.

	Seg A (1503)	Seg B (345)	Seg C (1242)
B 123	7.784%, 117	7.826%, 27	2.173%, 27

in the corresponding segments across the evaluation set. The value following the WER is the total number of errors.

Key finding: For the model finetuned on B123, the WER on **Value segments** is very low—significantly lower than the overall average WER. This is particularly advantageous for avionics setup, where accurate value extraction is critical. We examine each of the remaining errors in a later subsection.

2) *WER on Semantic Segments Involved in Action/Value Pair Matching:* When action/value pairs are extracted from an ATC instruction, pilots are expected to read back the critical information for verification purposes. This readback can be used for value matching to confirm the information extraction.

Table~V presents the segmented WER results for the same semantic categories (callsign, action, value) on **Class 2 transmissions** (pilot readbacks). Here, we provide results only from the model finetuned on B123.

As shown, the WER on pilot readbacks is notably higher than on ATC transmissions. We attribute this performance gap to several factors:

- **Lower receive quality:** Aircraft transmissions are typically received with weaker signal strength than ground-based ATC transmissions due to the location of data collection receivers (often situated at airports).
- **Data imbalance:** The audio recordings contain many more individual pilots than ATC controllers. Consequently, the training data provides substantially more examples per ATC controller than per pilot,

Table VI: WER on segments not used in VAAO.

	Seg F (38)	Seg G (368)	Seg H (136)
B 123	23.684%, 9	2.174%, 8	5.147%, 7

leading to poorer generalization to pilot voices.

Mitigation in real-world VAAO applications: Both issues can be partially addressed in operational VAAO systems. First, the VAAO system is onboard, which takes pilot’s audio directly from the audio panel. Second, when a VAAO system is deployed on a specific aircraft with a known set of pilots, the system can be finetuned individually for each pilot, adapting to their voice characteristics and readback patterns over time.

3) *WER on Semantic Segments Not Used in VAAO:* For comparison, we also report segmented WER on semantic segments that are not directly used for VAAO action/value extraction. Table~VI presents results for the following categories on Class 1 transmissions (ATC instructions), using the model finetuned on B123: F (conditions), G (facilities), and H (Nav aids).

Observation: The WER for Segment F (conditions) is substantially higher than the overall average. This is primarily due to the scarcity of training samples containing conditional phrases, which occur less frequently in the ATC0 dataset than callsigns, actions, and values.

4) *Key Insight: Critical Segments Achieve Lower WER:* Although the detailed segmented WER methodology is reserved for a separate paper, we report a striking finding in this work. Across multiple finetuned models, we consistently observe that **Value segments achieve substantially lower WER than the overall average WER**. In some configurations, Value WER is about one tenth of the overall WER.

This finding has significant practical implications for VAAO. It suggests that the ASR model preserves numerical and value-bearing information better than conversational content. This is plausible because values (altitudes, speeds, headings) are typically pronounced clearly and follow predictable patterns, whereas conversational elements are more variable. For avionics operation, where value extraction is paramount, this is excellent news: the ASR model naturally

performs best on the most critical parts of the transmission.

F. Turn Detection with Different Finetuned Models

As noted earlier, one of the primary motivations for using time-shift (cue-stitching) augmentation is to enable speaker turn detection within the finetuned ASR model. It would be remiss not to report the results.

Table VII shows the turn detection error rate, which is calculated as $(FT + FN) / (\text{Total Turns})$.

Table VIII shows the turn detection rate, which is calculated as $TP / (TP + FN)$.

Key observations:

- **Audio quality independence:** Turn detection performance is not strongly dependent on audio quality level. This is because audio quality labels are often determined by a few unintelligible words within a transmission. While quality affects ASR word error rate, turn detection relies on different acoustic and temporal cues, and is therefore less sensitive to the same degradations.
- **Batch processing focus:** The results reported here are for batch (offline) processing. A systematic study of turn detection performance in streaming scenarios with different latency requirements will be investigated elsewhere.
- **Comparison to other approaches:** Compared to dedicated speaker diarization systems or BERTraffic [citation], our approach offers the advantage of a single model for both transcription and turn detection. However, dedicated diarization may perform better in high-overlap or very noisy conditions. A direct comparison is left for future work.

1) *Answer to Question 4: Q-4:* Will speaker turn detection work effectively with models finetuned using turn-aware concatenation augmentation?

A-4: Yes. The finetuned model reliably outputs // tokens at speaker boundaries, achieving high precision and recall on held-out test data containing multiple speaker turns. As shown in Tables~VII and~VIII, both Augmentation approaches yield strong turn detection performance.

VII. CONCLUDING REMARKS

This paper presented a framework for finetuning ATC ASR models for voice-assisted avionics operations. Through quality-based data stratification (C-1), time-shift augmentation for joint transcription and speaker turn detection (C-2), and functional classification and semantic segmentation for fine-grained evaluation (C-3), we improved both model performance and our ability to assess it critically. Key findings include that Value segments achieve substantially lower WER than average WER, and that augmentation significantly benefits low-data regimes. Future work will integrate the downstream NLP unit for complete action/value extraction and extend the framework to streaming cockpit applications.

VIII. ACKNOWLEDGEMENT

Placeholder here. To be included later.

APPENDIX A. EXCERPT OF ERROR CASES FOR ACTIONS

Placeholder here. To be included later.

APPENDIX B. ERROR CASES FOR ACTION VALUES

Now, let's take a look at **all the 12 cases** of value errors for Class 1, Seg C.

We see that there is no error for dca_d1_1 and dca_d1_4.

The results for discrepancies for dca_f1_1 is shown below, where we show begin time in milliseconds, ASR, and ground truth (with differences in *italic*), and effect.

- 1093870: “two hundred *and* ten” vs “two hundred ten”. No effect.
- 1121000: “one seven *draw* knots” vs “one seven *zero* knots”. Number error.
- 1584060: “one seven zero *nasten*” vs “one seven zero *knots*”. Unit error.
- 1630700: “a hundred and *seven* till” vs “a hundred and *seventy* till”. Number error.
- 1932040: “one one niner *part* one” vs “one one niner *point* one”. Number error.
- 2054290: “a hundred and seventy *now*” vs “a hundred and seventy *knots*”. Unit error.
- 2298980: “a hundred and seventy *nine*” vs “a hundred and seventy *knots*”. Unit error.
- 2424250: “two hundred ten knots” vs “two hundred *and* ten knots”. No effect.
- 3117990: “one one niner *four* one” vs “one one niner *point* one”. Number error.

For dca_f2_2 (ASR vs Ground Truth)

- 653170: “*one* one niner point seven” vs “one niner point seven”. No effect.
- 692360: “*one* one niner point one” vs “one niner point one”. No effect.
- 1128530: “*one* one niner point one” vs “one niner point one”. No effect.

A few notes:

- In the 12 errors, we have 5 no-effect errors, 4 number errors, and 3 unit errors.
- We have checked the 3 no-effect errors in dca_f2_2; it is beyond the authors capability to definitely make sure that the ATC was saying “one niner point one” or “one one niner point one”.
- We have also checked the 2 no-effect errors in dca_f1_1. There is a slight trace of “and” in the first case, and there is strong evidence of “and” in the second case.
 - These are just the good examples of communication issues due to fast ATC speech rate.
 - These are also good examples of the robustness of the finetuned ASR.

Table VII: Turn detection error rate comparison using different augmentation approaches for finetuning.

	B1:L1	B1:L12	B1:L123	B123:L1	B123:L12	B123:L123
Aug 1	0.94%	1.04%	1.12%	0.70%	0.77%	0.94%
Aug 2	0.41%	0.40%	0.48%	0.48%	0.53%	0.79%

Table VIII: Turn detection rate comparison using different augmentation approaches for finetuning.

	B1:L1	B1:L12	B1:L123	B123:L1	B123:L12	B123:L123
Aug 1	0.94%	1.04%	1.12%	0.70%	0.77%	0.94%
Aug 2	0.41%	0.40%	0.48%	0.48%	0.53%	0.79%

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